

GRINNELL, IA, USA

WMO: 725292

Lat: 41.717N

Lon: 92.700W

Elev: 307

StdP: 97.69

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 14976

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -21.1	(c) -18.2	(d) -24.9	(e) 0.4	(f) -20.5	(g) -22.4	(h) 0.5	(i) -17.4	(j) 14.3	(k) -6.6	(l) 12.7	(m) -6.7	(n) 4.2	(o) 310	(p) 0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	32.7	23.8	31.2	23.1	30.0	22.4	25.6	30.2	24.6	29.2	23.7	28.3	4.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 24.1	(b) 19.8	(c) 29.0	(d) 22.9	(e) 18.4	(f) 27.5	(g) 22.3	(h) 17.6	(i) 26.8	(j) 80.8	(k) 29.9	(l) 75.8	(m) 29.3	(n) 72.3	(o) 28.3	(p) 28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.8	10.5	9.0	-25.7	34.7	3.4	1.9	-28.1	36.1	-30.1	37.2	-32.0	38.2	-34.5	39.6
			-26.0	27.3	3.3	0.8	-28.3	27.9	-30.2	28.4	-32.1	28.8	-34.4	29.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	-6.2	-4.3	3.1	10.0	16.7	21.9	23.5	22.0	18.7	11.4	3.8	-2.8	(6)
	DBStd	11.68	7.05	6.84	6.65	5.39	4.84	3.06	3.07	2.95	4.33	5.06	5.80	6.03	(7)
	HDD10.0	1844	503	403	231	66	6	0	0	0	1	43	197	396	(8)
	HDD18.3	3584	761	635	474	254	90	6	2	4	47	222	436	654	(9)
	CDD10.0	1802	0	1	16	64	213	357	417	373	263	87	11	1	(10)
	CDD18.3	501	0	0	1	3	38	112	161	118	59	7	0	0	(11)
	CDH23.3	4351	0	0	10	53	349	968	1494	929	482	65	1	0	(12)
	CDH26.7	1376	0	0	1	9	99	317	527	267	144	12	0	0	(13)
(14) Wind	WSAvg	4.2	4.9	4.8	4.9	5.3	4.6	3.8	2.9	2.6	3.2	4.0	4.7	4.5	(14)
(15) Precipitation	PrecAvg	946	27	32	58	96	123	139	112	121	82	69	52	35	(15)
	PrecMax	1361	55	76	129	196	262	309	295	333	180	161	137	111	(16)
	PrecMin	675	12	4	5	35	23	24	13	20	24	13	2	1	(17)
	PrecStd	206	9	20	39	45	48	65	68	78	41	49	35	26	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	17.1	24.9	27.9	32.6	33.1	35.4	33.2	32.7	28.0	21.9	14.1	(19)
		MCWB	7.3	11.4	15.9	17.2	21.5	23.7	25.1	23.3	22.9	18.9	13.9	11.9	(20)
	2%	DB	7.2	11.2	20.0	24.8	29.3	32.0	32.8	31.3	30.4	25.1	17.8	10.2	(21)
		MCWB	4.5	7.5	13.9	15.6	19.3	23.4	24.8	23.8	22.3	17.2	12.6	7.8	(22)
	5%	DB	4.8	7.4	16.3	22.0	27.2	30.3	31.2	29.7	28.1	22.5	15.0	7.7	(23)
		MCWB	3.0	4.8	11.6	14.1	18.4	21.7	24.1	23.1	21.0	16.4	10.1	5.6	(24)
	10%	DB	2.7	4.4	12.7	19.1	25.0	28.4	29.8	27.8	26.3	20.0	12.4	4.8	(25)
		MCWB	1.5	2.6	8.7	12.6	17.6	20.8	23.2	22.0	19.6	14.8	8.6	3.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.9	12.1	17.5	19.1	22.9	26.3	27.4	26.3	24.4	21.6	15.5	11.9	(27)
		MCDB	11.3	16.1	22.6	25.5	29.0	31.6	31.9	31.2	29.9	26.2	19.2	12.9	(28)
	2%	WB	4.8	7.9	15.2	17.1	21.3	24.7	26.1	25.0	23.4	19.7	13.3	8.6	(29)
		MCDB	6.9	10.9	18.0	22.3	27.1	29.8	30.4	29.3	28.6	22.3	17.3	9.4	(30)
	5%	WB	3.0	5.1	11.9	15.4	20.2	23.4	25.0	23.9	22.3	17.6	10.9	5.4	(31)
		MCDB	4.6	7.4	15.5	20.8	25.4	28.3	29.4	28.2	26.7	20.8	14.2	7.5	(32)
	10%	WB	1.6	2.7	9.0	13.7	19.0	22.3	24.0	23.0	21.2	15.5	8.9	3.0	(33)
		MCDB	2.7	4.4	12.5	18.2	23.2	26.9	28.4	27.0	24.6	19.4	11.8	4.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	8.8	9.4	11.4	10.9	10.4	10.3	10.2	11.4	10.9	9.6	7.9	(35)
		MCDBR	10.3	12.2	14.6	15.5	13.2	12.2	11.5	11.6	12.5	13.8	13.5	10.9	(36)
	5% WB	MCWBR	8.1	8.8	8.6	8.3	6.2	5.7	5.2	5.7	5.7	7.1	8.5	8.3	(37)
		MCDWR	9.9	12.2	12.8	13.4	11.3	10.7	10.1	9.9	10.7	11.0	12.3	10.6	(38)
(40) Clear-Sky Solar Irradiance	taub		0.286	0.305	0.330	0.378	0.408	0.419	0.435	0.414	0.383	0.339	0.299	0.285	(40)
		taud	2.408	2.360	2.388	2.275	2.262	2.280	2.217	2.293	2.332	2.452	2.492	2.430	(41)
	Ebn at Noon		868	902	918	891	869	858	838	846	849	853	849	836	(42)
		Edh at Noon	79	97	107	129	134	132	139	125	112	88	73	71	(43)
(44) All-Sky Solar Radiation	RadAvg		1.90	2.92	3.81	4.73	5.56	6.16	6.23	5.42	4.57	3.04	2.05	1.56	(44)
	RadStd		0.18	0.21	0.37	0.36	0.41	0.60	0.43	0.33	0.34	0.36	0.18	0.14	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page